

Unrolled List
–Destroy List
–Program Analysis

Program

```
void destroyList() {
    if (head == NULL) {
        cout << "The list is already empty.\n";
        return;
    }

    struct Node* temp = head;
    while (temp != NULL) {
        struct Node* nextNode = temp->next;
        free(temp);
        temp = nextNode;
    }

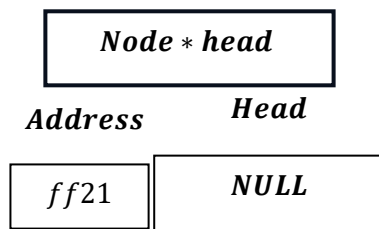
    head = NULL;
    cout << "The entire list has been destroyed and memory
freed.\n";
}

...
    case 5:
        destroyList();
        break;
```

Program Analysis

1. List is Empty

```
if (head == NULL) {  
    cout << "The list is empty.\n";  
    return;  
}
```

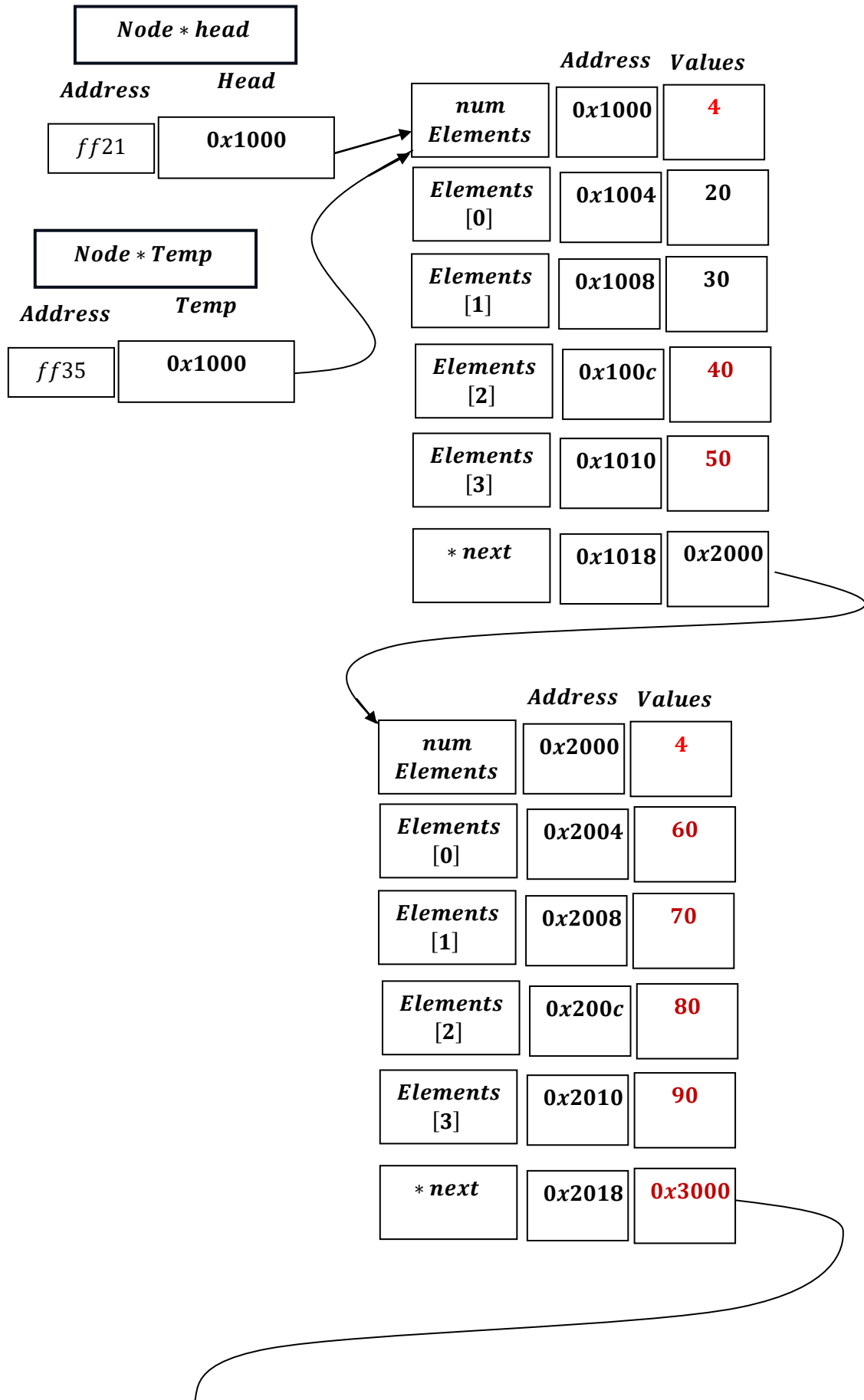


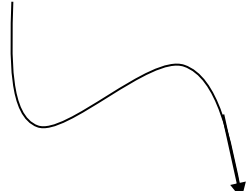
As head = NULL, it prints : `` The list is empty.``

Return void and exit from the function.

2. List is not Empty

```
struct Node* temp = head;
```

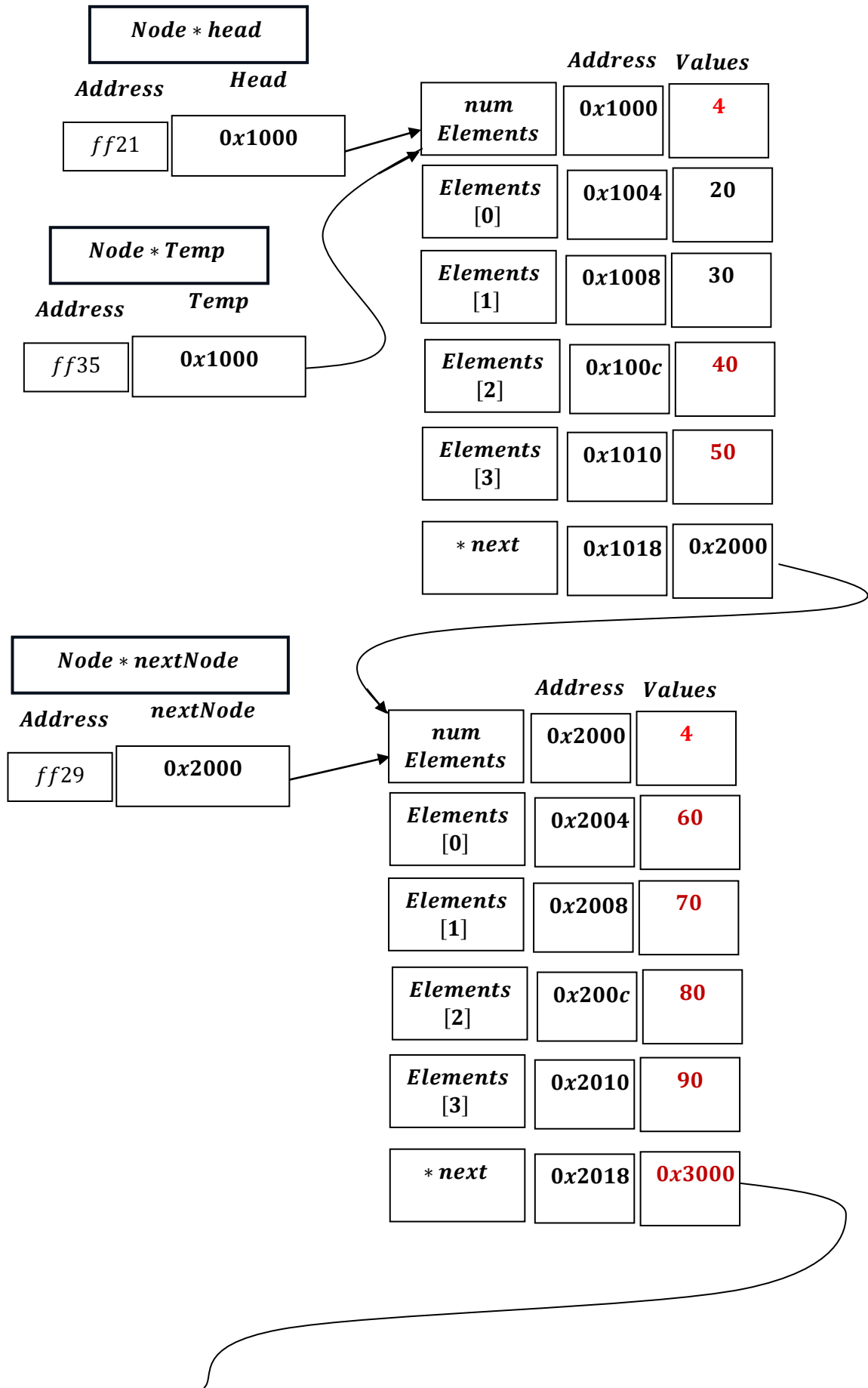


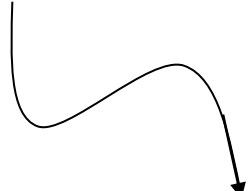


	<i>Address</i>	<i>Values</i>
<i>num Elements</i>	0x3000	4
<i>Elements</i> [0]	0x3004	100
<i>Elements</i> [1]	0x3008	110
<i>Elements</i> [2]	0x300C	120
<i>Elements</i> [3]	0x3010	130
<i>* next</i>	0x3018	NULL

```
while (temp != NULL) {
    struct Node* nextNode = temp->next;
    free(temp);
    temp = nextNode;
}
```

$While(temp: 0x1000 \neq NULL) \left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right]$
 $struct Node * nextNode = temp \rightarrow next: 0x2000$





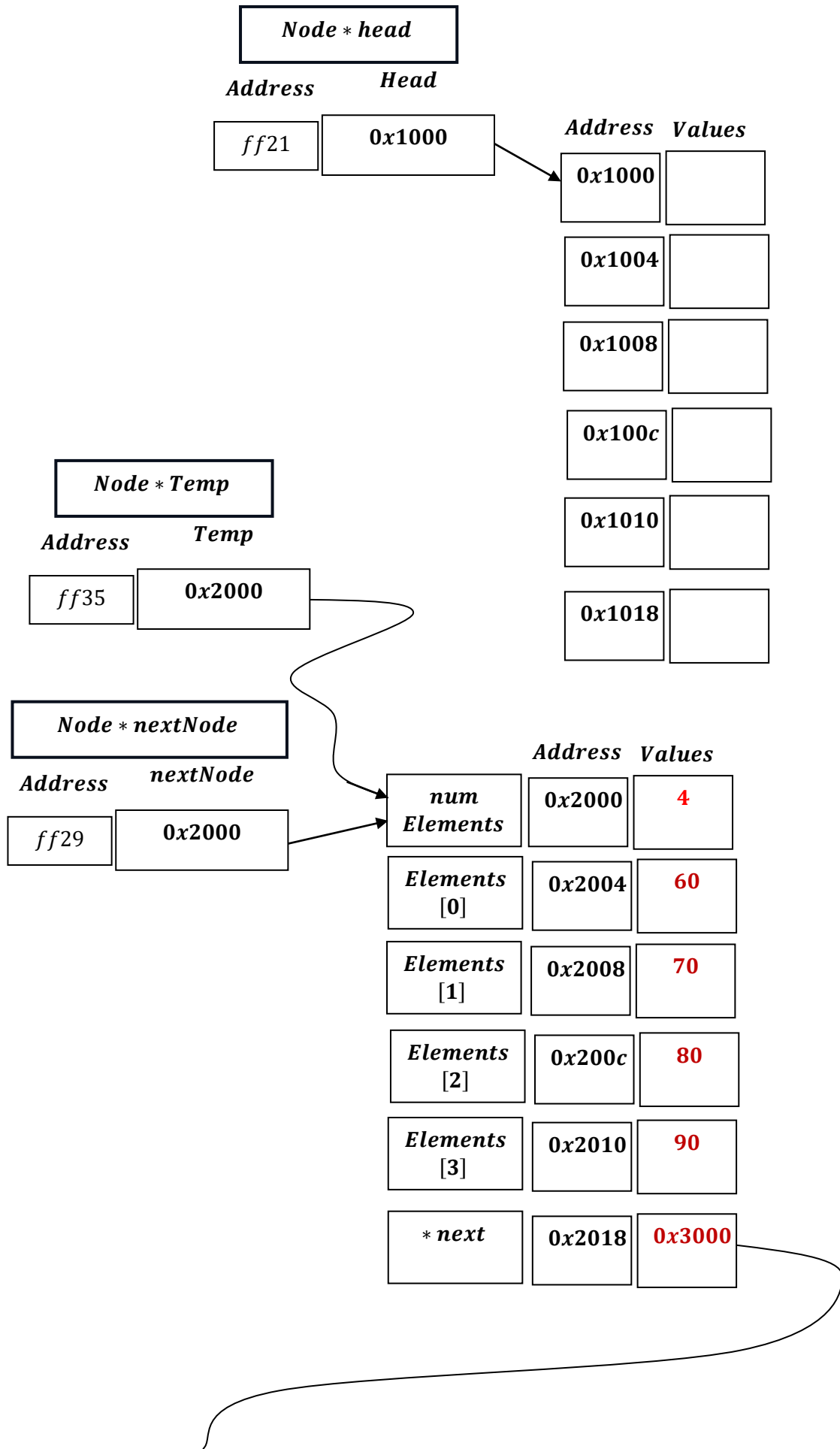
	<i>Address</i>	<i>Values</i>
<i>num Elements</i>	0x3000	4
<i>Elements [0]</i>	0x3004	100
<i>Elements [1]</i>	0x3008	110
<i>Elements [2]</i>	0x300C	120
<i>Elements [3]</i>	0x3010	130
<i>* next</i>	0x3018	NULL

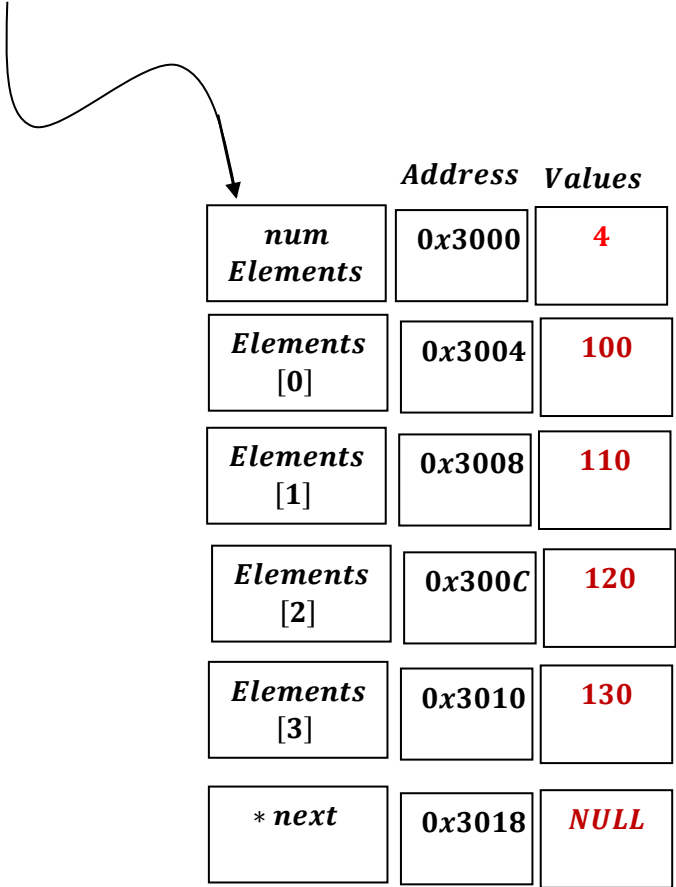
That is:

```

While(temp: 0x1000  $\neq$  NULL)  $\left[ \begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right]$ 
    struct Node * nextNode = temp  $\rightarrow$  next: 0x2000
    free(temp: 0x1000)
    temp = nextNode = 0x2000
  
```

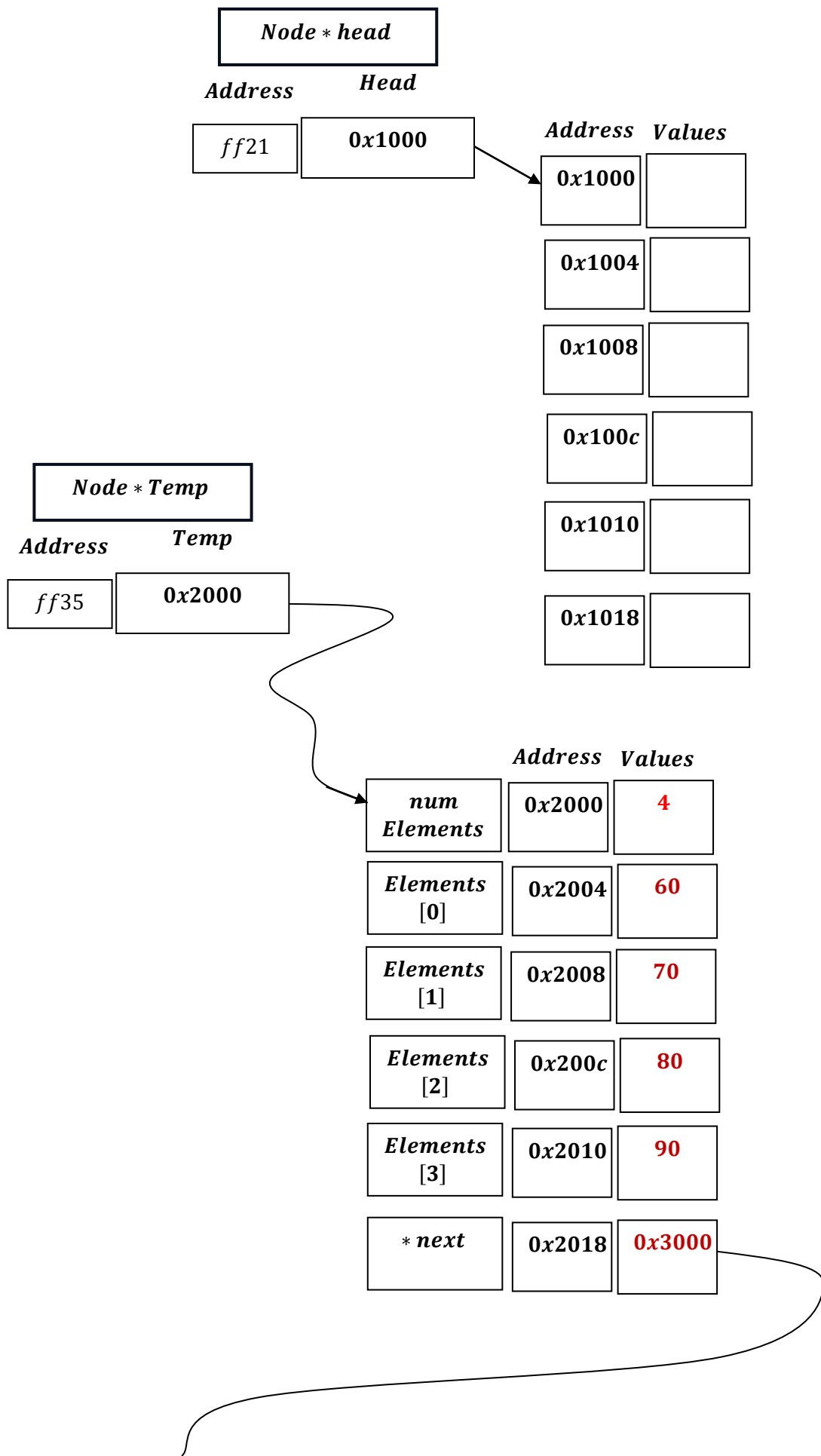
*Note: **free(temp: 0x1000)**, this will freeing memory : **0x1000** starting from **0x1000** and its associated memory allocations ended till : **0x1018** are freed all at once .*

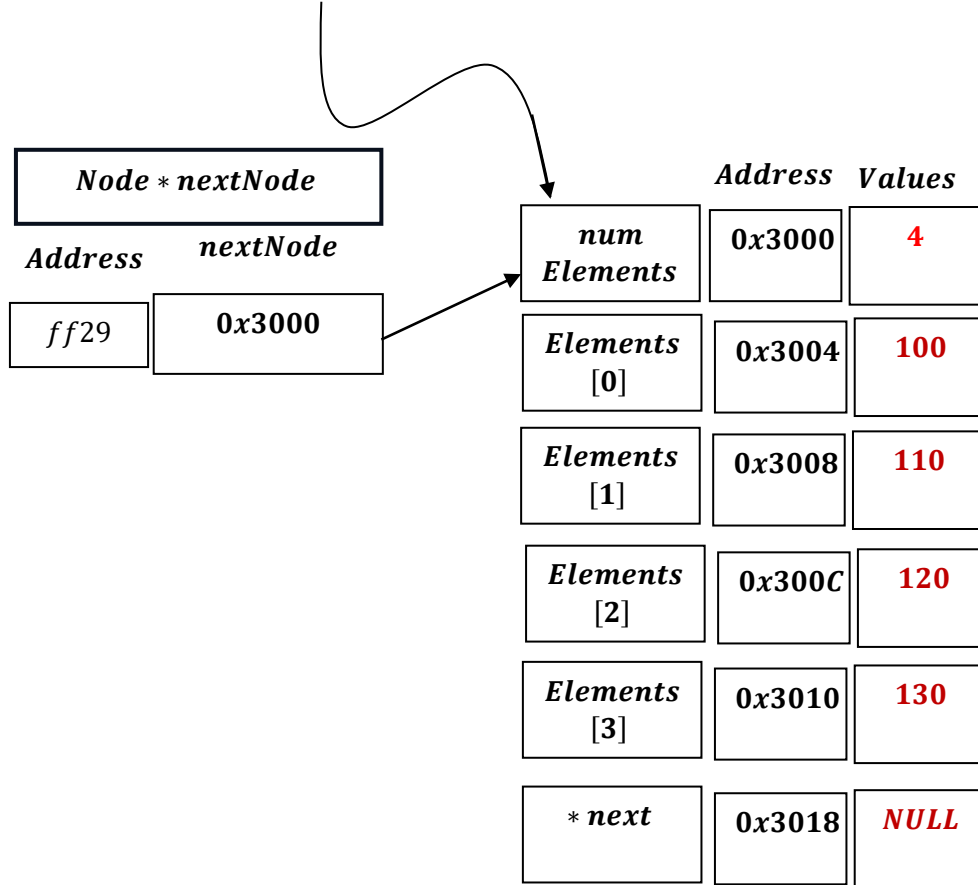




	<i>Address</i>	<i>Values</i>
<i>num Elements</i>	0x3000	4
<i>Elements</i> [0]	0x3004	100
<i>Elements</i> [1]	0x3008	110
<i>Elements</i> [2]	0x300C	120
<i>Elements</i> [3]	0x3010	130
<i>* next</i>	0x3018	NULL

$While(temp: 0x2000 \neq NULL) \left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right]$
 $struct Node * nextNode = temp \rightarrow next: 0x3000$



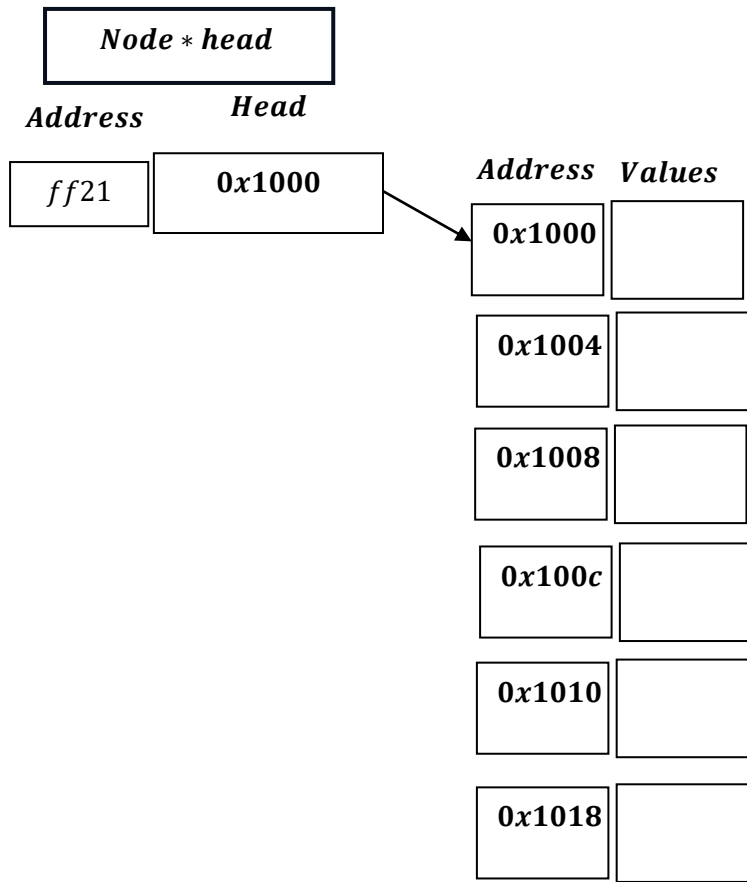


That is:

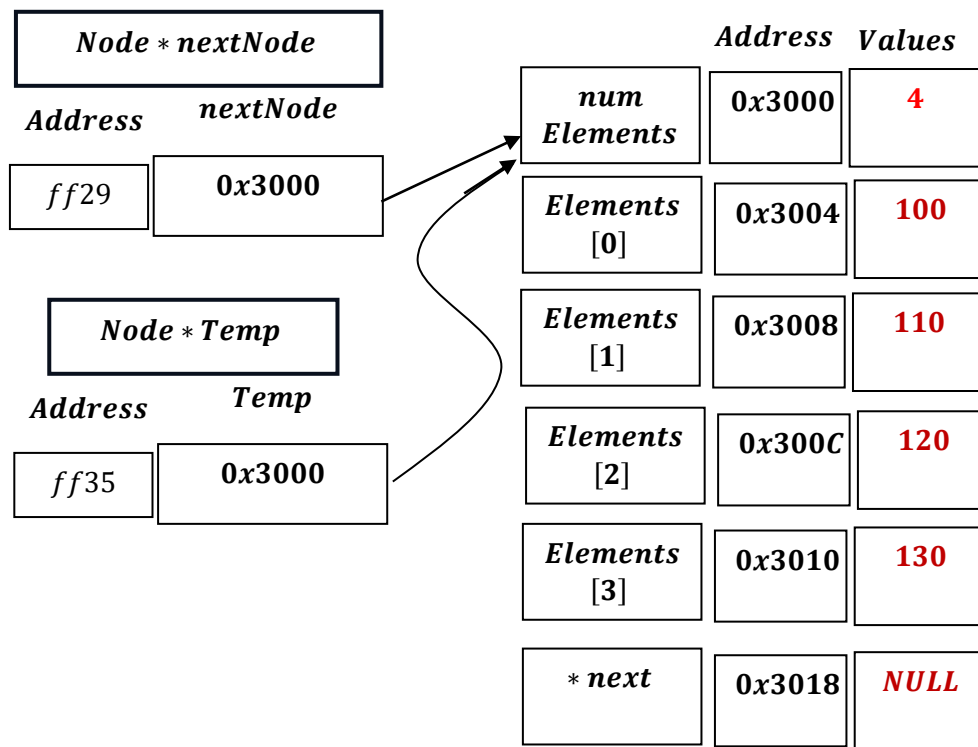
```

While(temp: 0x2000 ≠ NULL)  $\left[ \begin{array}{c} \text{Condition} \\ \text{is} \\ \text{True} \end{array} \right]$ 
    struct Node * nextNode = temp → next: 0x3000
    free(temp: 0x2000)
    temp = nextNode = 0x3000
  
```

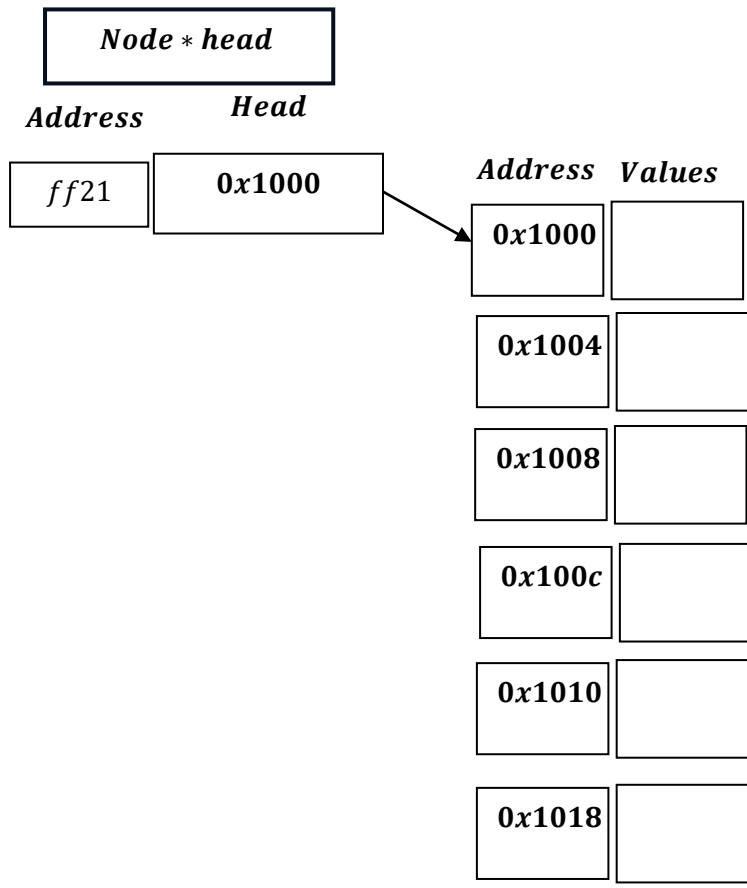
*Note: **free(temp: 0x2000)**, this will freeing memory : **0x2000** starting from **0x2000** and its associated memory allocations ended till : **0x2018** are freed all at once .*



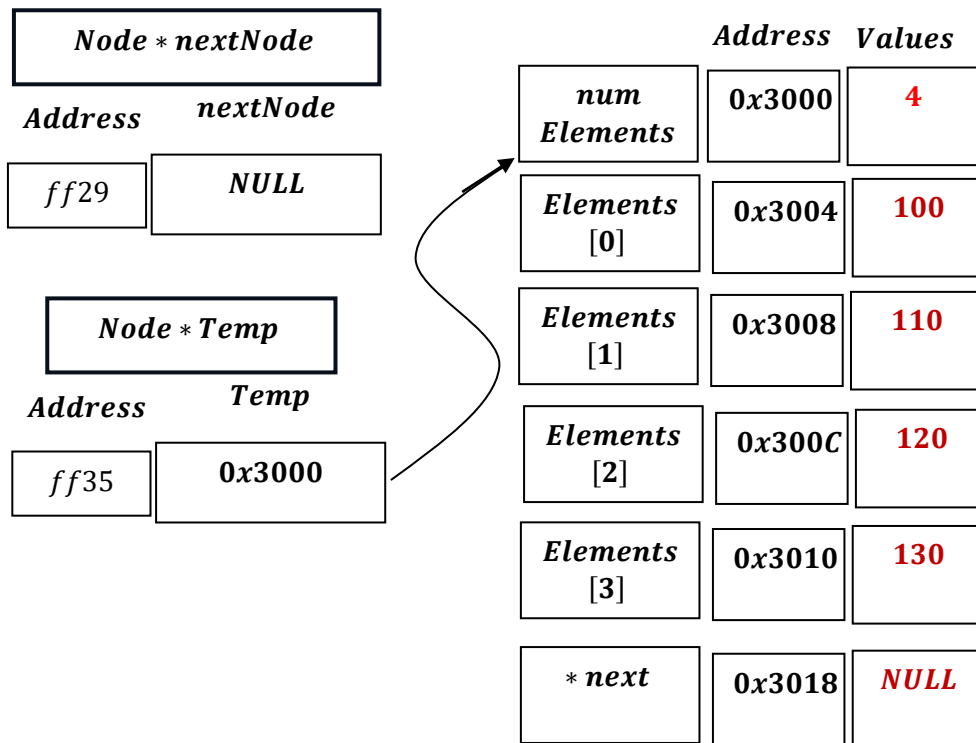
Address	Values
0x2000	
0x2004	
0x2008	
0x200c	
0x2010	
0x2018	



While(*temp: 0x3000* $\neq$ *NULL*) $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right]$
 *struct Node * nextNode = temp* $\rightarrow$ *next: NULL*



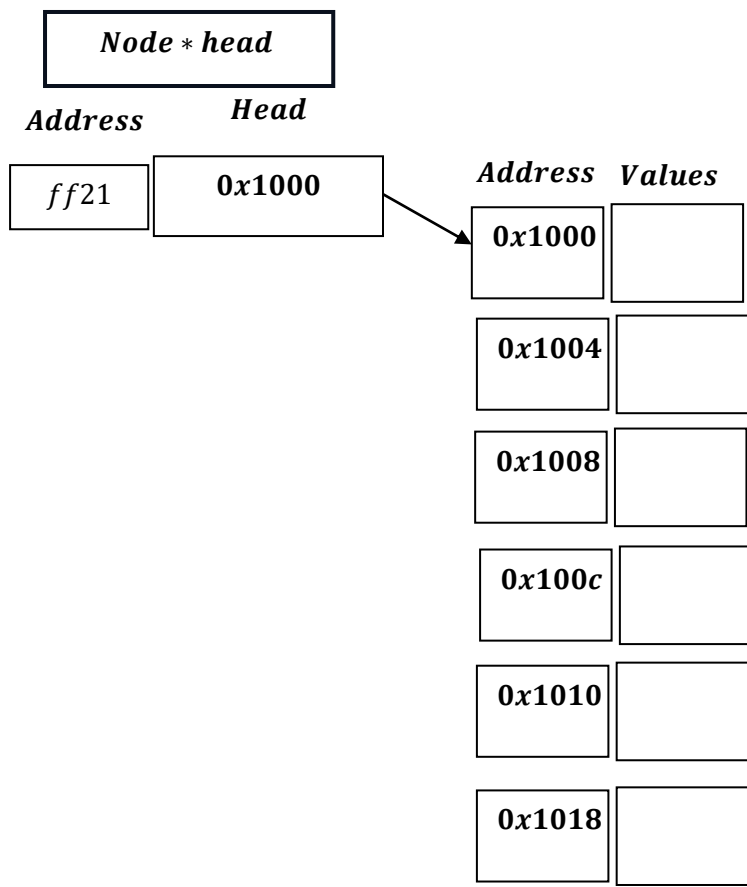
Address	Values
0x2000	
0x2004	
0x2008	
0x200c	
0x2010	
0x2018	



That is:

*While(temp: 0x3000 $\neq$ NULL) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{True} \end{array} \right]$
 *struct Node *nextNode = temp $\rightarrow$ next: NULL*
 free(temp: 0x3000)
 *temp = nextNode = NULL**

*Note: **free(temp: 0x3000)**, this will freeing memory : **0x3000** starting from **0x3000** and its associated memory allocations ended till : **0x3018** are freed all at once .*



Address	Values
0x2000	
0x2004	
0x2008	
0x200c	
0x2010	
0x2018	

Node * nextNode

Address nextNode

<i>ff29</i>	<i>NULL</i>
-------------	--------------------

Node * Temp

Address Temp

<i>ff35</i>	<i>NULL</i>
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Address Values

<i>0x3000</i>	
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<i>0x3004</i>	
----------------------	--

<i>0x3008</i>	
----------------------	--

<i>0x300C</i>	
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<i>0x3010</i>	
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<i>0x3018</i>	
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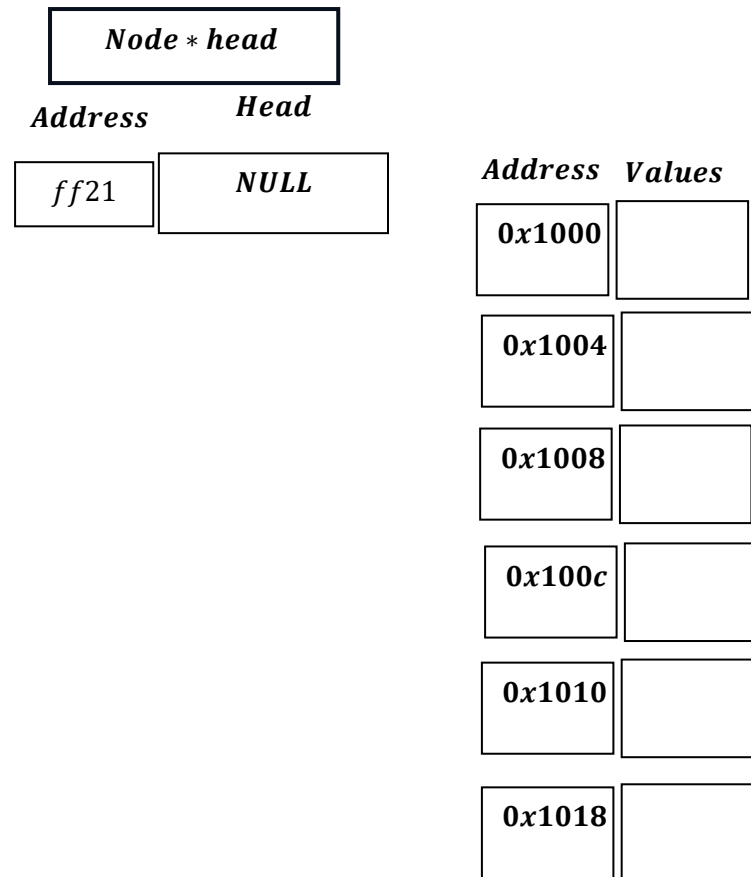
Now , While(temp: NULL $\neq$ NULL) $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{False} \end{array} \right]$

As, condition is false – while loop exits.

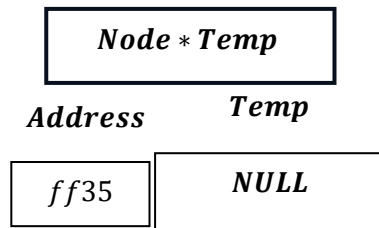
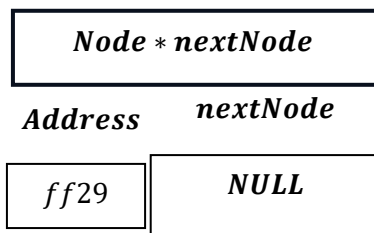
```

    head = NULL;
    cout << "The entire list has been destroyed and memory
freed.\n";

```



Address	Values
0x2000	
0x2004	
0x2008	
0x200c	
0x2010	
0x2018	



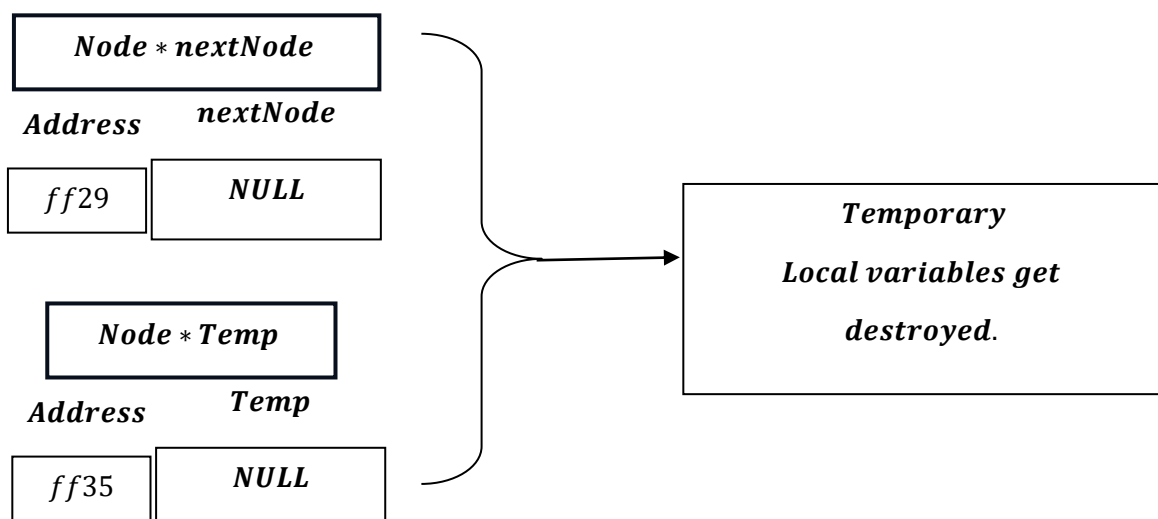
Address	Values
0x3000	
0x3004	
0x3008	
0x300C	
0x3010	
0x3018	

And now it will print:

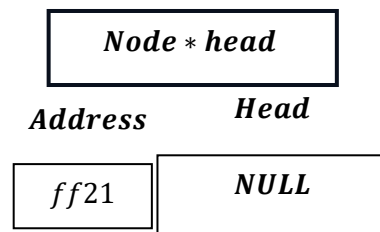
```
cout << "The entire list has been destroyed and memory
freed.\n";
```

Print: ``The entire list has been destroyed and memory freed.``

As , functions ends :



And we are left with:



That is list is empty.
